# Measurement and

# Sensor Technologies

03

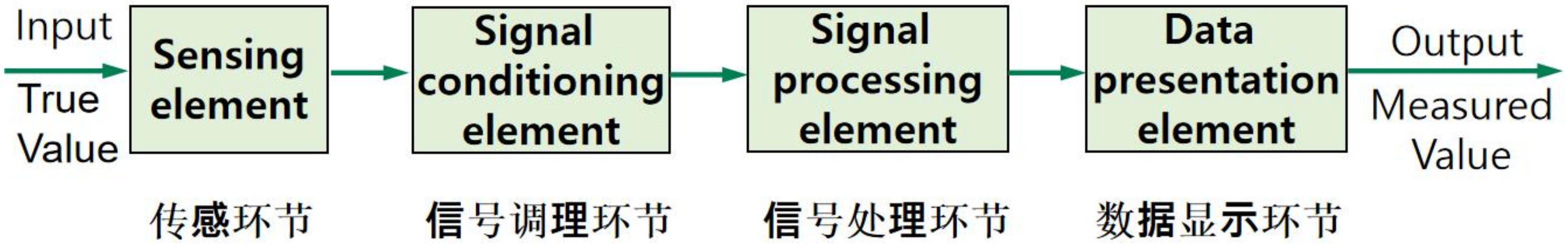
# Signal Conditioning

Signal conditioning can include amplification, filtering, transformation, conversion, modulation, range matching, isolation and many more processes which may be required to make sensor output suitable for processing after conditioning.

# 3 Signal Conditioning

- 3.1 Introduction of Signal Conditioning
- 3.2 Filtering
- 3.3 Modulation
- 3.4 Summary

# 3.1 Introduction of Signal Conditioning




- In the study of electronics, signal conditioning signifies manipulating an analog signal in such a way that it meets the requirements of the next stage for further processing.
- Signal conditioning can include amplification, **filtering**, transformation, conversion, **modulation**, range matching, isolation and many more processes which may be required to make sensor output suitable for processing after conditioning.

# 3.2 Filtering

## 1) What the filter does

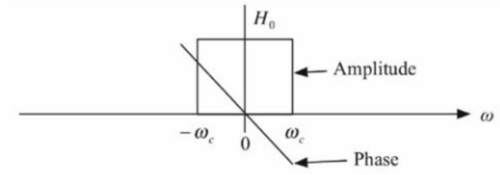
- A filter is a frequency selection device that allows certain frequency components in a signal to pass while greatly attenuating other frequency components

- ✓ Pass specific frequency components in the signal
- ✓ Greatly attenuates other frequency components

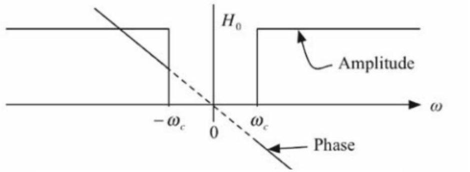
——Filters out interfering noise or unwanted frequency band signals

## 2) Types of filter

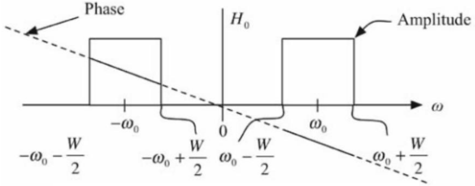
■ The filters are specified based on a transfer function $H(j\omega)$ in terms of its amplitude $|H(j\omega)|$, phase $\angle H(j\omega)$, or delay responses– $|\mathrm{d}\angle H(j\omega)/\mathrm{d}\omega|$.



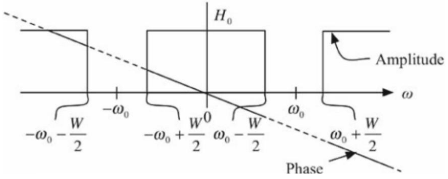

Low-pass




High-pass




Band-pass




Band elimination

# 3.2.1 Simple Low-pass Filters

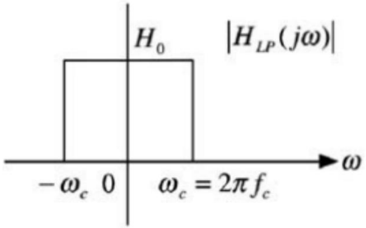
## 1) The ideal low-pass filter transfer function

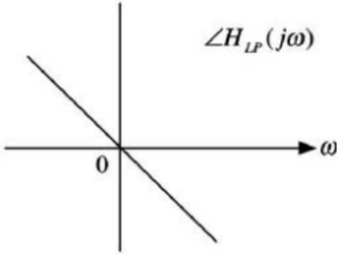
■ The words low-pass means that when the signalis $x(t)$ passed through a low-pass filter, only low frequencies, say 0 to $f_c = \omega_c/2\pi$, are passed and block all frequencies above the cutoff frequency $f_c$.

$$H_{LP}(j\omega) = H_0 \prod \left[\frac{\omega}{2\omega_c}\right] e^{-j\omega t_0}$$

$$|H_{LP}(j\omega)| = H_0 \prod \left[\frac{\omega}{2\omega_c}\right] = \begin{cases} H_0 & |\omega| \leq \omega_c \\ 0 & |\omega| \geq \omega_c \end{cases}, \omega_c = 2\pi f_c$$

$$\angle H_{LP}(j\omega) = -\omega t_0$$



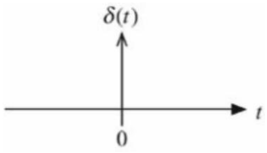

■The phase response is assumed to be linear, i.e., slope is constant. The group delay is $t_0 = -d\frac{\angle H_{LP}(j\omega)}{\mathrm{d}\omega}$ .

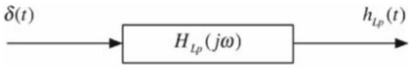
# 3.2.1 Simple Low-pass Filters

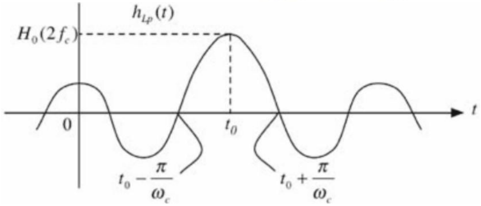
## 2) The impulse response of the ideal low-pass filter

■ For a physically realizable system, the impulse response $h(t) = 0$ for $t < 0$, i.e., the system is causal. For a realizable system, the output cannot exist before the input is applied. The impulse response of the ideal low-pass filter can be determined from below.

$$h_{Lp}(t) = F^{-1}[H_{LP}(j\omega)] = F^{-1}\left[H_0 \prod\left[\frac{\omega}{2\omega_c}\right] e^{-j\omega t_0}\right] = H_0(2f_c)\frac{\sin\left(\omega_c(t-t_0)\right)}{\omega_c(t-t_0)}$$



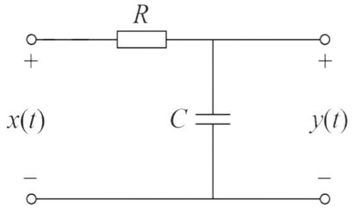




■There is output before the input is applied. The ideal low-pass filter is not causal and is physically unrealizable.

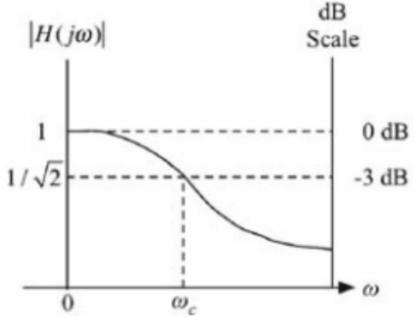
# 3.2.1 Simple Low-pass Filters

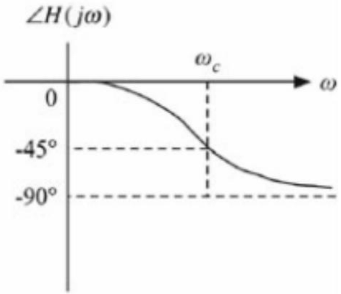
## 3) RC low pass filter

■ The RC filter circuit is simple, and the standard resistance-capacitance components are easily available, so it is often used in the field of engineering testing. The circuit of the RC low-pass filter and its amplitude-frequency and phase-frequency characteristics are shown in the following figure：




$$H_{LP}(j\omega) = \frac{1}{(1 + j\omega RC)}$$




■Cut-off frequency $f_c = f_{3dB} = (1/2\pi RC)$ ，it is shown that the RC value determines the cutoff frequency of the filter.

# 3.2.1 Simple Low-pass Filters

**Example 3.1**

■ Consider the RC circuit shown in Figure 3.6 with the source and the load resistors. Derive the transfer function and sketch the amplitude characteristic function for the two cases. a.$R_L = \infty$ and b.$R_L = R_S$.

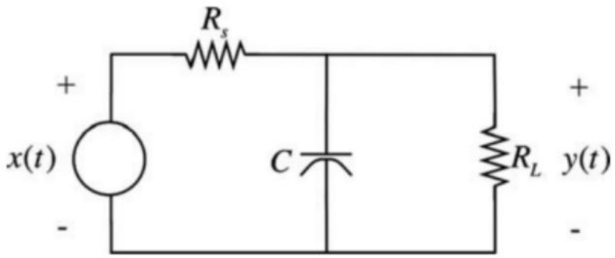



Figure 3.6 Example 3.1

# 3.2.1 Simple Low-pass Filters

**Example 3.1**

■ Solution: The transfer functions are given by

$$\left[\frac{Y(s)}{X(s)}\right] = H_{LP}(s) = \frac{Z_2}{Z_2 + Z_1} \qquad Z_2 = \frac{R_L/Cs}{R_L + \frac{1}{Cs}} = \frac{R_L}{1 + R_L Cs}, \; Z_1 = R_s$$

$$\left[\frac{Y(s)}{X(s)}\right]_b = H_{LP}(s) = \frac{\frac{R_L}{R_s R_L C}}{s + \left[\frac{R_L + R_s}{R_s R_L C}\right]} = \frac{K}{s + \omega_c} \qquad K = \frac{1}{R_s C}, \omega_c = \frac{R_s + R_L}{R_s R_L C}$$

■In case a, the load resistance is infinite, i.e., the circuit is not loaded. In case b. $R_L = R_S$. For the two cases the corresponding transfer functions are

$$\text{a. } H_{LP,R_L\to\infty}(s) = \frac{1/R_s C}{s + \frac{1}{R_s C}} \qquad \text{b. } H_{LP,R_L=R_S}(s) = \frac{1/R_s C}{s + \frac{2}{R_s C}}$$

# 3.2.1 Simple Low-pass Filters

**Example 3.1**

a. $H_{LP,R_L\to\infty}(s) = \dfrac{1/R_sC}{s+\dfrac{1}{R_sC}}$ b. $H_{LP,R_L=R_S}(s) = \dfrac{1/R_sC}{s+\dfrac{2}{R_sC}}$

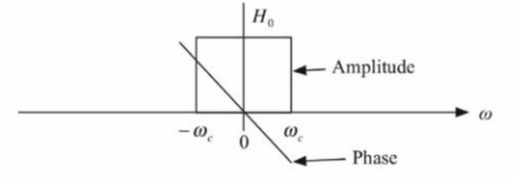
■In both cases, the gain constant is the same. However, the cutoff frequency is increased in the case of a load resistance. Note that the peak value of the amplitude response function in case b. is 1/2. So, the 3dB frequency corresponds to the value of the magnitude of the function equal to $(1/2)\times(1/\sqrt{2})$. At $\omega=0$ the filter circuit is transparent; at $\omega=\infty$ there is no signal transmission; in between these frequencies, the output signal amplitude attenuation is determined by the equation $|Y(j\omega)| = |H(j\omega)||X(j\omega)|$. At the 3dB frequency $\omega_{3dB}$.

$$|Y(j\omega)|_{\omega=\omega_{3\mathrm{dB}}} = \frac{1}{\sqrt{2}}|X(j\omega)|_{\omega=\omega_{3\mathrm{dB}}}$$

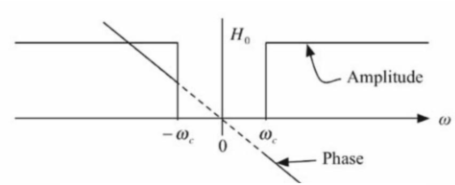
# 3.2.2 Simple High-Pass Filters

## Generation from low-pass filtering to high-pass filtering

■ In the ideal low- and high-pass filter cases, it can be see that




Low-pass




High-pass

$$|H_{LP}(j\omega)|_{\omega=0} = |H_{HP}(j\omega)|_{\omega=\infty} = 1 \text{ and}$$

$$|H_{HP}(j\omega)|_{\omega=0} = |H_{LP}(j\omega)|_{\omega=\infty} = 0$$

$$1(\text{low}-\text{pass}) \to 0(\text{high}-\text{pass}) \text{ or}$$

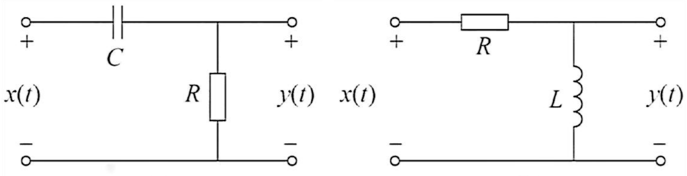
$$0(\text{high}-\text{pass}) \to 1(\text{low}-\text{pass})$$

■A logical conclusion is that $\omega \to 1/\omega$ (i.e.,$s \to 1/s$) provides a transformation that gives a way to find a high-pass filter function from a low-pass filter function.

# 3.2.2 Simple High-Pass Filters

## Generation from low-pass filtering to high-pass filtering

■ Using the RC and the RL low-pass circuits, we have two simple high-pass filters shown in Figure 3.7 a, b, one is a RL and the other one is a RC circuit.



(a) (b)

$$H_{Hpa}(s) = \frac{s}{s+\frac{1}{RC}}, H_{Hpb}(s) = \frac{s}{s+\frac{R}{L}}$$

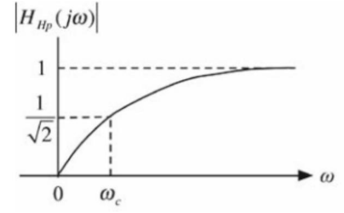
$$H_{Hpa}(j\omega) = \frac{j\omega}{j\omega+\frac{1}{RC}}, H_{Hpb}(j\omega) = \frac{j\omega}{j\omega+\frac{R}{L}}$$

$$|H_{Hpa}(j\omega)| = \frac{|\omega|}{\sqrt{\omega^2+(\frac{1}{RC})^2}}$$

$$\angle H_{Hpa}(j\omega) = 90^\circ - \tan^{-1}(\omega RC)$$

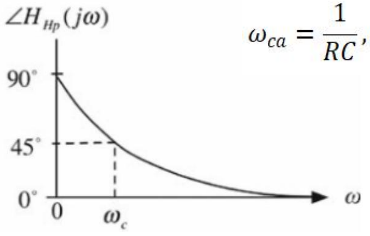
$$|H_{Hpb}(j\omega)| = \frac{|\omega|}{\sqrt{\omega^2+(\frac{R}{L})^2}}$$

$$\angle H_{Hpb}(j\omega) = 90^\circ - \tan^{-1}\left(\frac{\omega L}{R}\right)$$



(a)

$$\omega_{ca} = \frac{1}{RC}, \omega_{cb} = \frac{R}{L}$$



(b)

# 3.2.3 Simple Band-Pass Filters

## 1) Second order transfer function of a bandpass filter

■ A second-order function that has the band-pass characteristics is

$$H_{Bp}(s) = \frac{H_0 as}{s^2 + as + b} = \frac{H_0 \frac{\omega_0}{Q} s}{s^2 + \frac{\omega_0}{Q} s + \omega_0^2}$$

- The center frequency $\omega_0 = \sqrt{\omega_{\text{low}}\omega_{\text{high}}}$ , $\omega_{\text{low}}$ and $\omega_{\text{high}}$ are two cutoff (or 3 dB) frequencies;
- The 3 dB bandwidth $\beta = \omega_{\text{high}} - \omega_{\text{low}}$ ;
- The quality factor $Q = \frac{\omega_0}{\omega_{\text{high}} - \omega_{\text{low}}}$ ;

$\omega_0$ is not in the middle of the pass-band. It is the geometric center of the pass-band edges.

■For simplicity the gain constant is assumed to be $H_0 = 1$ in the following. The corresponding transfer function, the amplitude, and phase responses are given by

$$H_{Bp}(j\omega) = \frac{(\frac{\omega_0}{Q})j\omega}{\omega_0^2 - \omega^2 + j(\frac{\omega_0}{Q})\omega} = \frac{j}{Q(\frac{\omega_0}{\omega} - \frac{\omega}{\omega_0}) + j}$$

$$|H_{Bp}(j\omega)| = \frac{1}{\sqrt{1 + Q^2(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})^2}}$$

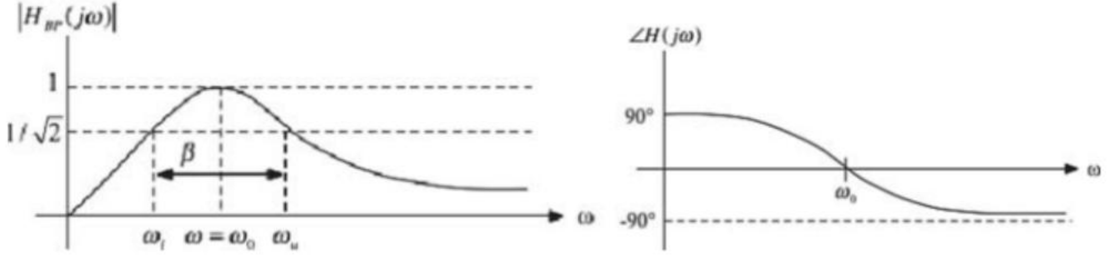
$$\angle H_{Bp}(j\omega) = \tan^{-1} Q(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})$$

# 3.2.3 Simple Band-Pass Filters

## 1) Second order transfer function of a bandpass filter

$$|H_{Bp}(j\omega)| = \frac{1}{\sqrt{1+Q^2\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)^2}}$$

$$\angle H_{Bp}(j\omega) = \tan^{-1} Q\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)$$




(a) (b)

$$|H_{Bp}(j\omega)|^2\Big|_{\omega=\omega_l,\omega_u} = \frac{1}{2} \rightarrow Q^2\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)^2 = 1$$

Assuming that $Q > 1/2$ ,the positive roots are given by $\omega_l\,,\omega_u = \omega_0\sqrt{1+\frac{1}{4Q^2}} \mp \frac{\omega_0}{2Q}$

$$\beta = \omega_{\text{high}} - \omega_{\text{low}} = \frac{\omega_0}{Q} \qquad \omega_0^2 = \omega_l\omega_u \rightarrow \omega_0 = \sqrt{\omega_l\omega_u}$$

The bandwidth decreases as $Q$ increases and vice versa. The filter is assumed to be narrowband if $\omega_0$ is very large compared to the bandwidth of the filter, i.e. As a rough measure, we assume the filter is a narrowband filter if $Q = \frac{\omega_0}{\beta} \geq 10$.For the narrowband case the edges of the pass-band and $\omega_0$ are given below.

$$\omega_l\,,\omega_u = \omega_0 \mp \frac{\omega_0}{2Q}$$

In this case, $\omega_0$ is in the middle of the 3 dB frequencies and is the center frequency.

# 3.2.3 Simple Band-Pass Filters

## Example 3.2

■ Consider the circuit shown in Figure 3.10. Find the transfer function and derive the expressions for the center frequency, bandwidth, and the quality factor.

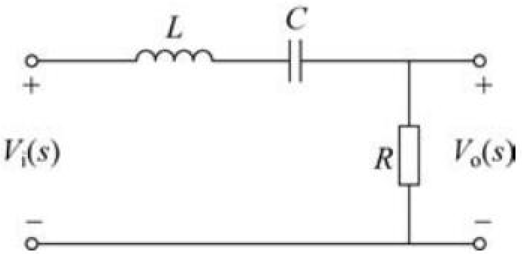



Figure 3. 10 A simple band-pass filter

■ Solution: The transfer functions are

$$H_{Bp}(s)=\frac{v_0(s)}{v_i(s)}=\frac{\frac{R}{L}s}{s^2+\frac{R}{L}s+\frac{1}{LC}} \qquad \left|H_{Bp}(j\omega)\right|=\frac{\frac{R}{L}|\omega|}{\sqrt{(\frac{1}{LC}-\omega^2)^2+(\omega\frac{R}{L})^2}}$$

$$H_{Bp}(j\omega)=\frac{\frac{R}{L}j\omega}{\frac{1}{LC}-\omega^2+j\omega\frac{R}{L}} \qquad \angle H_{Bp}(j\omega)=90°-\tan^{-1}\frac{\frac{R}{L}\omega}{\frac{1}{LC}-\omega^2}$$

$$\omega_0=\sqrt{\frac{1}{LC}},\ \beta=\frac{R}{L} \qquad Q=\frac{\omega_0}{B}=\sqrt{\frac{L}{CR^2}}$$

# 3.2.3 Simple Band-Pass Filters

**Example 3.2**

■ Also, inductors are never ideal and can be modeled by a resistor in series with an ideal inductor shown in Figure 3.11 resulting in the impedance of the nonideal inductor as $R_i + sL$. The new transfer function and the amplitude response are

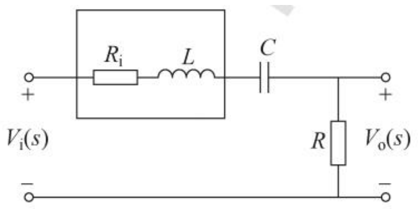



Figure 3.11 A simple band-pass filter with a nonideal inductor

$$H_{Bp}(s) = \frac{\frac{R}{L}s}{s^2 + \frac{R+R_i}{L}s + \frac{1}{LC}}$$

$$|H_{Bp}(j\omega)| = \frac{\frac{R}{L}|\omega|}{\sqrt{(\frac{1}{LC} - \omega^2)^2 + \omega\frac{R_i + R}{L}}}$$

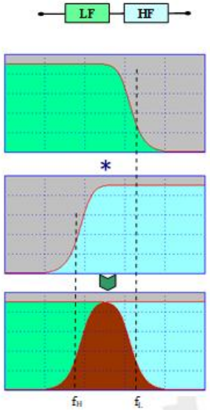
- The center frequency is the same as before.
- The maximum value of the amplitude response is now $R/(R_i + R)$.
- The new bandwidth is $(R_i + R)$/L.
- The Q value is reduced.

Nonideal inductors make the amplitude response less peaked with an increase in the bandwidth, i.e., the amplitude response becomes lower and broader.

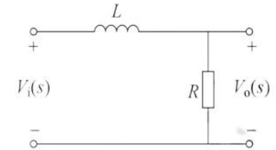
# 3.2.3 Simple Band-Pass Filters

## 2) Construction of a Bandpass Filter

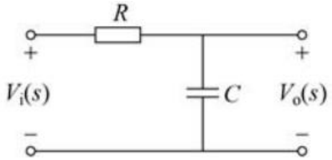
■ A band-pass filter can be viewed as a cascaded low-pass and a high-filter combination with the cutoff frequency of the low-pass filter greater than the cutoff frequency of the high-pass filter, $\omega_{c,Lp} > \omega_{c,Hp}$.




low pass filter



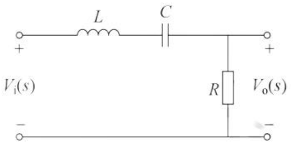

low pass filter



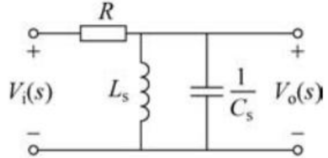

replace L by a series LC circuit

replace C by a parallel LC circuit

bandpass filter




bandpass filter




In the two simple low-pass circuits considered earlier, the RL circuit, with the replacement of the inductor by a series LC circuit results in the circuit studied above. By replacing the capacitor in the RC low-pass circuit by a parallel LC circuit results in a band-pass circuit.

# 3.2.4 Simple Band-Elimination Filters

## Second-Order Transfer Function of Band-Stop Filter

■ A second-order notch filter has a transfer function of the form

$$H_{Be}(s) = \frac{s^2 + 2bs + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

A special and an interesting case is when $b = 0$ and for this case

$$\lim_{s\to 0} H_{Be}(s) = 1, \lim_{s\to\infty} H_{Be}(s) = 1$$

$$|H_{Be}(j\omega)| \cong \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + \left(\frac{\omega_0}{Q}\omega\right)^2}} \qquad \angle H_{Be}(j\omega) = -\tan^{-1}\frac{\frac{\omega_0}{Q}\omega}{\omega_0^2 - \omega^2}$$

$$|H_{Be}(j\omega)|_{\omega=0} = 1$$

$$\lim_{\omega\to\infty} |H_{Be}(j\omega)| = 1$$

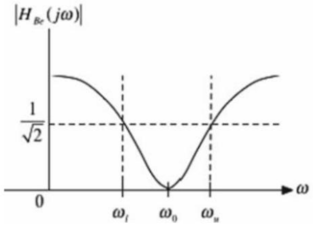
- The 3 dB frequencies are obtained by equating $|H(\omega_{3dB})| = 1/\sqrt{2}$. The two frequencies are

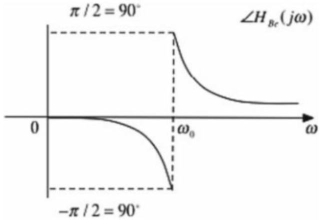
$$\omega_l, \omega_u = \quad \omega_0 \pm \frac{\omega_0}{2Q}, \beta = \frac{\omega_0}{Q}$$

If $b \neq 0$, then $|H_{Be}(j\omega)|_{\omega=\omega_0} = \frac{Qb}{\omega_0}$

- The attenuation will be significant at the notch frequency as $\omega_0$ is usually large. Also, note the phase reversal at $\omega = \omega_0$ in the phase response.

Notch filters are used wherever a narrowband of frequencies needs to be eliminated from a received signal.



# 3.2.4 Simple Band-Elimination Filters

## Example 3.3

■ Band-elimination filters can be derived from low-pass filters by replacing an inductor (capacitor) by a parallel LC (series LC) circuit. Consider the circuit in Figure 3.15 with a nonideal inductor with the equivalent impedance of the coil equal to $R_W + j\omega Ls$. Assuming the circuit is not loaded, the output current is zero. Derive the transfer function and show it corresponds to a band-elimination filter.

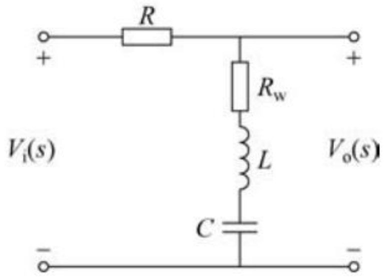



Figure 3.15 A band-elimination filter with a nonideal inductor

# 3.2.4 Simple Band-Elimination Filters

## Example 3.3

- Solution: The transfer function, its amplitude and phase responses are

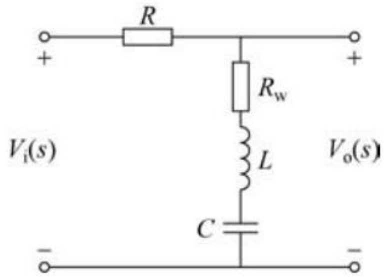



Figure 3.15 A band-elimination filter with a nonideal inductor

$$H_{Be}(s)=\frac{R_w+Ls+\frac{1}{Cs}}{R+R_w+Ls+\frac{1}{Cs}}=\frac{1+R_ws+LCs^2}{1+LCs+(R+R_w)Cs}$$

$$|H_{Be}(j\omega)|=\frac{\sqrt{(1-\omega^2LC)^2+(\omega R_wC)^2}}{\sqrt{(1-\omega^2LC)^2+(\omega(R+R_w)C)^2}}$$

$$H_{Be}(j\omega)=\frac{R_w+j\omega L+\frac{1}{j\omega C}}{R+R_w+j\omega L+\frac{1}{j\omega C}}=\frac{1-\omega^2LC+j\omega R_wC}{1-\omega^2LC+j\omega(R+R_w)C}$$

$$\angle H_{Be}(j\omega)=\tan^{-1}\frac{R_wC\omega}{1-\omega^2LC}-\tan^{-1}\frac{\omega(R+R_w)C}{1-\omega^2LC}$$

The transfer function has the band-elimination characteristics as

$$\lim_{\omega\to 0}H_{Be}(j\omega)=1\,,\ \lim_{\omega\to\infty}|H_{Be}(j\omega)|=1\ \text{ and }\ |H_{Be}(j\omega)|_{\omega=\frac{1}{\sqrt{LC}}}=\frac{R_w}{R+R_w}$$

If the winding resistance $R_W$ is large, then the peak value of the amplitude response is close to 1. we have $2b=R_W/L$, $\omega_0=1/\sqrt{LC}$ and $\omega_0/Q=(R+R_W)/L$. The notch bandwidth is $B=|\omega_0/Q|=|(R+R_W)/L|$. In the ideal case, a second-order notch filter can be obtained from a second-order band-pass filter by replacing the parallel (series) LC circuit by a series (parallel) LC circuit.

# 3.2 Filtering

(a)low pass filter

(b)high pass filter

(c)bandpass filter

(d)band stop filter

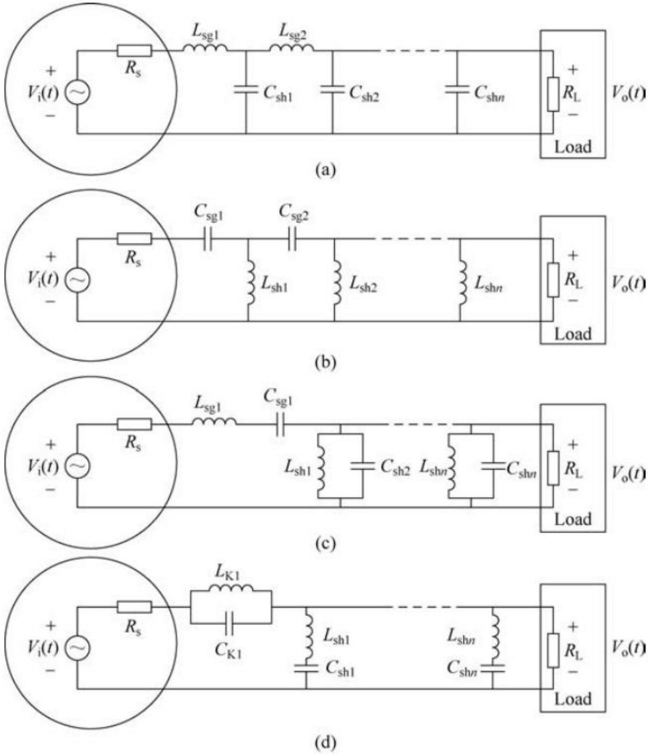
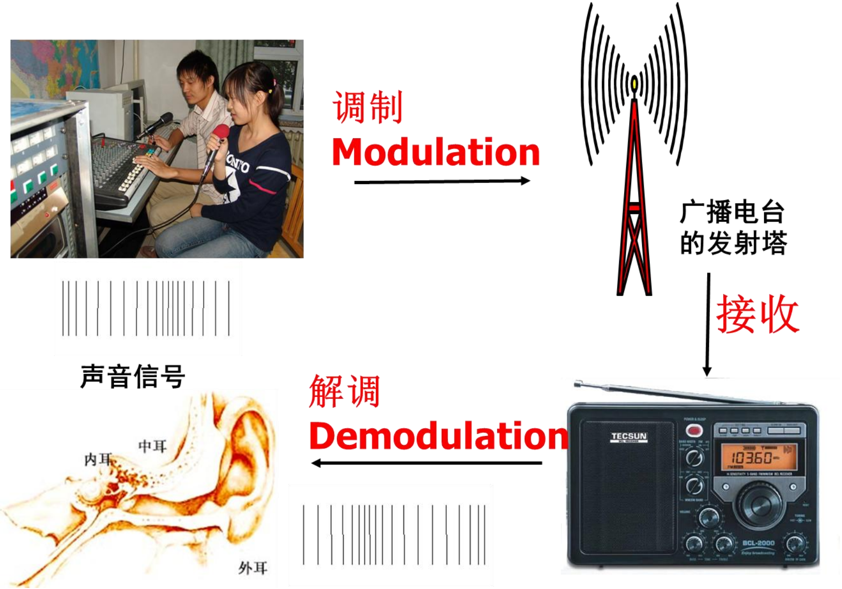



Figure 3.16 Passive filters

# 3.3 Modulation

## 1) Purpose of modulation

- To maintain realizable antenna height, to provide the facility of multiplexing and channel separation and to get higher level of noise immunity, a number of modulation schemes are employed both in the case of analog and digital communication.



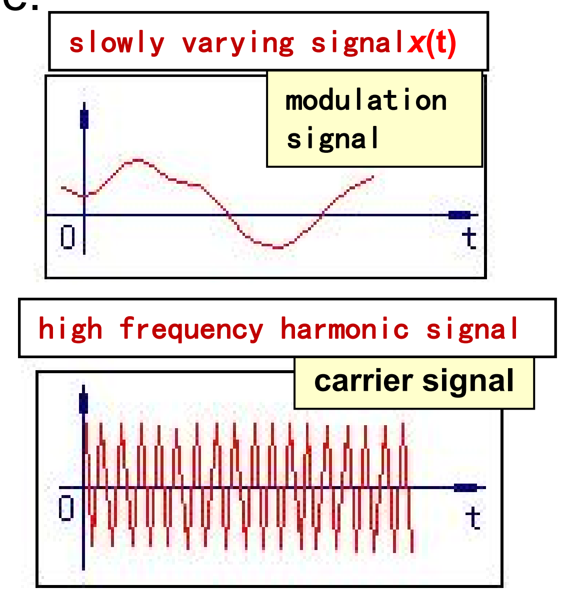

- **Realize the transmission of slowly varying signals, especially over long distances**
- **Improve anti-jamming capability and signal-to-noise ratio in signal transmission**

Control a certain parameter (amplitude, frequency or phase) of a high-frequency AC signal (carrier signal) with a slowly varying signal (modulation signal), then AC amplify, transmit, and finally demodulate the amplified signal

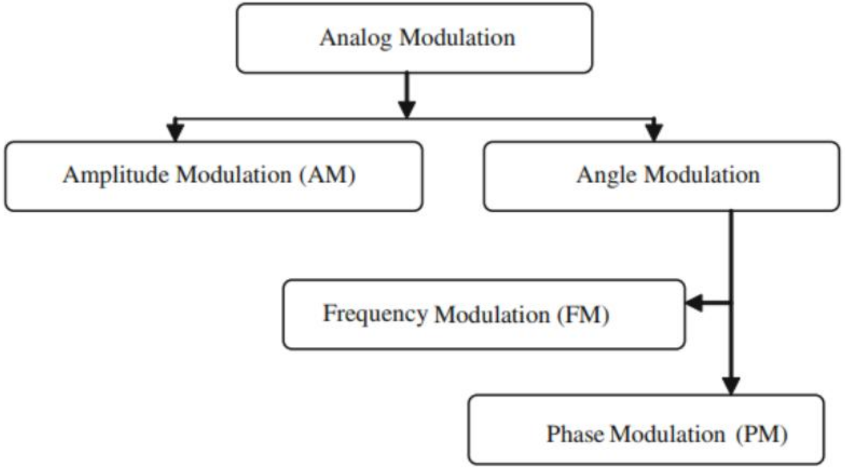
# 3.3 Modulation

## 2) Types of modulation

■ The classification is as shown in Figure.




$$y(t) = A_0 \cos(2\pi f_0 t + \phi_0)$$




**a) Amplitude Modulation幅值调制(简称调幅，AM)**

**b) Frequency Modulation频率调制(简称调频，FM)**

**c) Phase Modulation相位调制(简称调相，PM)**

# 3.31 Amplitude Modulation

## 1) Modulation Index(AM)

- The amplitude of the carrier signal is modulated according to the instantaneous amplitude of the modulating signal. In other words change in amplitude, is proportional to the modulating signal $m(t)$.By considering the proportionality constant 1.

$$\Delta\text{amplitud} \propto m(t) \Rightarrow \Delta\text{amplitud} = m(t)$$

- The modulation index for AM$(\mu)$ is inversely proportional to the amplitude of carrier signal$(V_c)$ and directly proportional to the amplitude of modulating signal$(V_m)$.

$$\mu \propto \frac{1}{V_c} \quad , \quad \mu \propto V_m \quad \Rightarrow \quad \mu = \frac{V_m}{V_c}$$

# 3.31 Amplitude Modulation

## 2) Math model of AM

- Modulating signal $m(t) = V_m \sin(\omega_m t)$ and single tone carrier $V_c(t) = V_c \sin(\omega_c t)$ where $\omega_c \gg \omega_m$.Modulated carrier would be

$$\varphi_{AM}(t) = A(t)\sin\omega_c t \quad \text{where,} \quad A(t) = V_c + \Delta\text{amplitud} = V_c + m(t) = V_c + V_m \sin\omega_m t$$

$$\varphi_{AM}(t) = (V_c + V_m \sin\omega_m t)\sin\omega_c t = V_c \sin\omega_c t + \frac{V_m}{2}[\cos(\omega_c - \omega_m)t - \cos(\omega_c + \omega_m)t]$$

$$\varphi_{AM}(t) = V_c \sin\omega_c t + \frac{\mu V_c}{2}\cos(\omega_c - \omega_m)t - \frac{\mu V_c}{2}\cos(\omega_c + \omega_m)t$$

- The amplitude modulated signal contains three frequency components as $f_c$, $f_c$+$f_m$ and $f_c$-$f_m$. The 2 said frequencies are named as lower and upper side frequency signals. The bandwidth of the single tone AM is $[(f_c + f_m) - (f_c - f_m)] = 2f_m$.

# 3.31 Amplitude Modulation

## 2) Math model of AM

**modulation signal:** $m(t) = V_m \sin \omega_m t$

**carrier signal:** $V_c(t) = V_c \sin \omega_c t$

**amplitude modulation wave:** $\varphi_{AM}(t)$

$$\varphi_{AM}(t) = V_c \sin \omega_c t + \frac{\mu V_c}{2} \cos(\omega_c - \omega_m)t - \frac{\mu V_c}{2} \cos(\omega_c + \omega_m)t$$

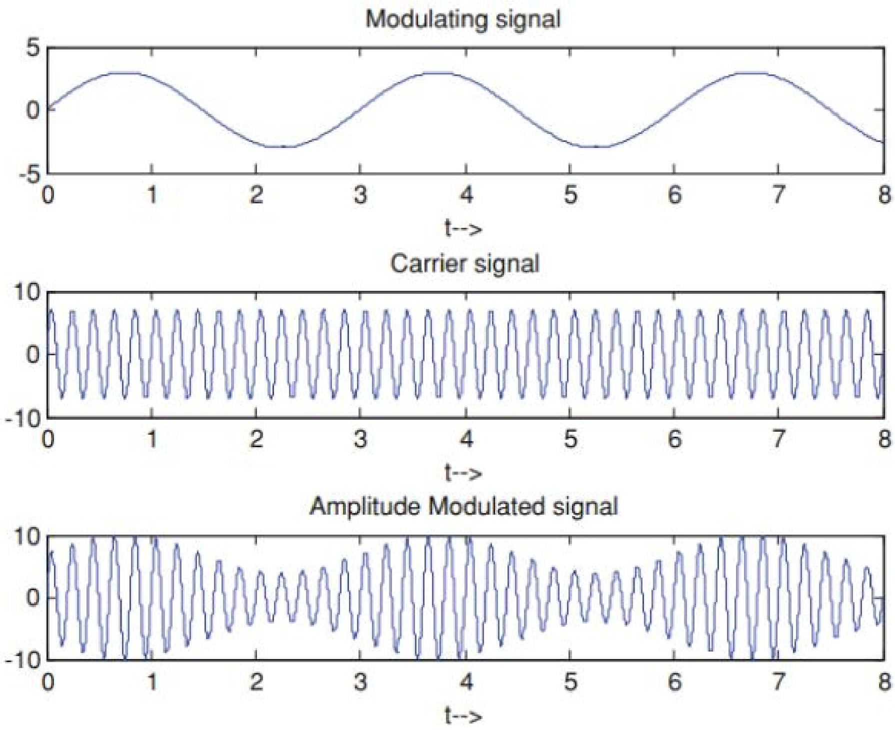



Figure 3.19 Waveforms Before and After Amplitude Modulation

# 3.32 Angle Modulation

## 1) Mathematic Model

■ We can modulate the angle or phase of a signal in two ways. As the angle of a signal is essentially a function of the frequency, we can modulate frequency (FM) or directly modulate the entire angle/phase (PM) to get angle modulation.

FM: Modulating signal $m(t)$ and carrier signal $V_c(t) = Acos(\omega_c t)$, The frequency after modulation is $\omega_c + \Delta\omega$.

$$\Delta\omega \propto m(t), \Delta\omega = k_f m(t)$$

where $k_f$ is the constant of frequency modulation. Therefore, the instantaneous frequency of carrier after modulation is

$$\omega_i = \omega_c + \Delta\omega = \omega_c + k_f m(t)$$

$$\theta_i = \int_{-\infty}^{t} \omega_i(t) = \int_{-\infty}^{t} [\omega_c + k_f m(t)]\, dt = \omega_c t + k_f \int_{-\infty}^{t} m(t)\, dt$$

frequency modulation wave: $\varphi_{FM}(t) = A\cos\theta_i(t) = A\cos\left[\omega_c t + k_f \int_{-\infty}^{t} m(t)\, dt\right]$

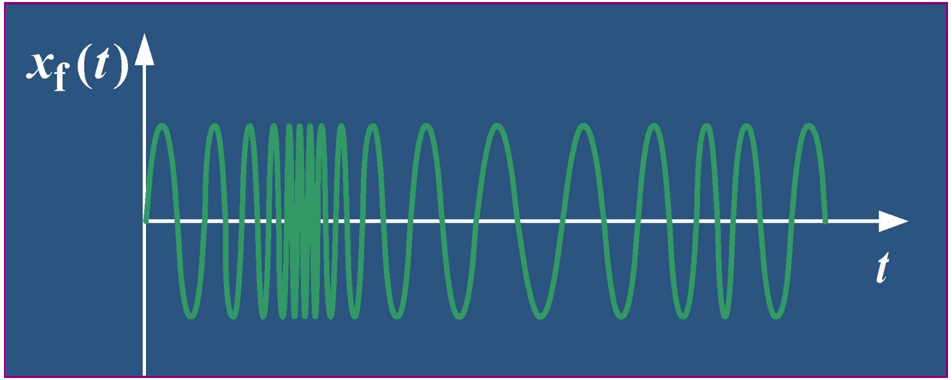
# 3.32 Angle Modulation

## 1) Mathematic Model

FM is to use the amplitude of the signal $m(t)$ to modulate the frequency of the carrier, or the FM wave is a constant-amplitude wave with different sparse density that changes with the voltage amplitude of the signal $m(t)$.

frequency modulation wave:

$$\varphi_{FM}(t) = A\cos\theta_i(t) = A\cos\left[\omega_c t + k_f \int_{-\infty}^{t} m(t)\,dt\right]$$




FM waves are equal amplitude waveforms that vary in density as the signal changes.

# 3.32 Angle Modulation

## 1) Mathematatic Model

■ Similarly, if we employ Phase Modulation, the change in angle is directly proportional to the modulating signal.

PM: Modulating signal $m(t)$ and carrier signal $V_c(t) = A\cos(\omega_c t)$, The angle after modulation is $\omega_c t + \Delta\theta$.

$$\Delta\theta \propto m(t),$$

$$\Delta\theta = k_p m(t)$$

where $k_f$ is the constant of phase modulation. Therefore, the instantaneous angle/phase of carrier after modulation is

$$\theta_i = \omega_c t + \Delta\theta = \omega_c t + k_p m(t)$$

$$\omega_i = \frac{\mathrm{d}\theta_i(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\omega_c t + k_p m(t)\right) = \omega_c + k_p \frac{dm(t)}{dt}$$

phase modulation wave: $\varphi_{PM}(t) = A\cos\theta_i(t) = \varphi_{FM}(t) = A\cos\left[\omega_c t + k_p m(t)\right]$

# 3.32 Angle Modulation

## 2) FM and PM are Interchangeable

■ we can change the entire angle (or phase) of carrier signal either by modifying the frequency or by modifying the phase. That's why PM and FM are the two sub-classes of the broader class of angle modulation. From the mathematical models derived in the previous section, it can also be shown that FM and PM are interchangeable and thereafter inseparable.

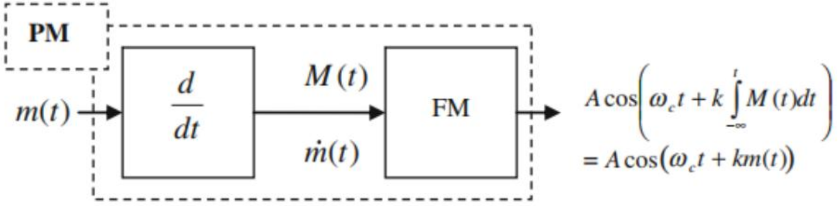



Figure 3.20 FM modulator along with a differentiator used as PM Modulator

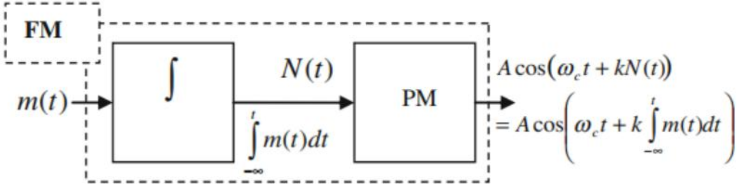



Figure 3.21 PM modulator along with an integrator used as FM modulator

## 3.4 Summary

■ The basic low-pass,high-pass,band-pass,band-elimination filters and the ideal filter functions that describe their functions,and simple circuits that can be used as filters have discussed.

■Analog modulation signifies modification of three basic properties of analog single tone carrier signal according to the analog base band signal.